

SPARC Galactic Kinematics under Conditional AQUAL/Picard Inputs Quarantined legacy scaffold—not for citation or submission

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I. INTRODUCTION

The rotation curves of disk galaxies exhibit a systematic discrepancy between Newtonian predictions from visible baryons and observed kinematic velocities in the low-acceleration regime ($a \lesssim 1.2 \times 10^{-10} \text{ m/s}^2$) [1, 2]. In standard Λ CDM cosmology, this discrepancy is addressed by postulating individual cold dark matter (CDM) halos whose density parameters (e.g., Navarro-Frenk-White scale radius r_s and characteristic density ρ_0) are freely adjusted on a galaxy-by-galaxy basis [3].

In the Integrated Toroidal-Syntropic Model (ITSM), low-acceleration dynamics emerge as the non-linear response of a finite-density superfluid vacuum condensate. In companion papers, the theoretical foundation is established:

1. **Paper I:** Present-epoch scale matching and the field-rescaling invariant $C_{\text{obs}} = C_m^{3/2}/\sqrt{C_{\text{IR}}}$ [4].
2. **Paper II:** Anisotropic Casimir stress and absence of free-field attractors on rectangular T^3 [5].
3. **Paper III:** Gate-structured multi-scale falsification program and Bogoliubov acoustic modes [6].

Here (Paper IV), we present the kinematic evaluation of the full SPARC database under the 2D/3D nonlinear AQUAL Picard solver without dark matter halos.

II. FIRST-PRINCIPLES FIELD EQUATIONS & NUMERICAL PICARD SOLVER

The static, weak-field action of the ITSM scale-compensator field ψ is given by [7, 8]:

$$S_\psi = - \int d^4x \left[\frac{f^2}{2} |\nabla\psi|^2 \mu \left(\frac{|\nabla\psi|}{\ell a_0} \right) + C_m \rho_b \psi \right], \quad (1)$$

where conformal Weyl invariance uniquely fixes $C_m \equiv 1.0$, and scale matching fixes $f = 1/\sqrt{4\pi G}$ and the transition healing length $\ell = \sqrt{4\pi G}/a_0 \approx 0.21 \text{ mm}$.

Varying with respect to ψ yields the nonlinear AQUAL Poisson equation:

$$\nabla \cdot \left[\mu \left(\frac{|\nabla\psi|}{\ell a_0} \right) \nabla\psi \right] = 4\pi G \rho_b(R, z). \quad (2)$$

We solve Eq. (2) on a 129×129 axisymmetric cylindrical (R, z) grid using Picard relaxation with under-relaxation parameter $\omega = 0.70$:

$$\nabla \cdot \left[\mu(|\nabla\psi^{(n)}|) \nabla\psi^{(n+1)} \right] = 4\pi G \rho_b. \quad (3)$$

Outer boundary conditions are computed via exact free-space logarithmic Green function convolution over boundary nodes, driving the truncation residual down to $\varepsilon = 6.06 \times 10^{-9}$ in 32 iterations.

III. SPARC DATABASE KINEMATIC BENCHMARKS

The SPARC catalog provides high-precision Spitzer $3.6 \mu\text{m}$ photometry and neutral hydrogen (HI) rotation curves for 175 disk galaxies spanning over five decades in stellar mass and gas fraction [2].

For each galaxy i , the total baryonic Newtonian velocity is:

$$V_{\text{bar}}^2(R) = |V_{\text{gas}}|V_{\text{gas}} + \Upsilon_{\text{disk}}|V_{\text{disk}}|V_{\text{disk}} + \Upsilon_{\text{bulge}}|V_{\text{bulge}}|V_{\text{bulge}}, \quad (4)$$

and the total rotation velocity is predicted by:

$$V_{\text{ITSM}}(R) = \sqrt{V_{\text{bar}}^2(R) \cdot \nu \left(\frac{V_{\text{bar}}^2(R)}{R a_0} \right)}, \quad (5)$$

where $\nu(y) = \frac{1}{2} + \frac{1}{2}\sqrt{1 + 4/y}$.

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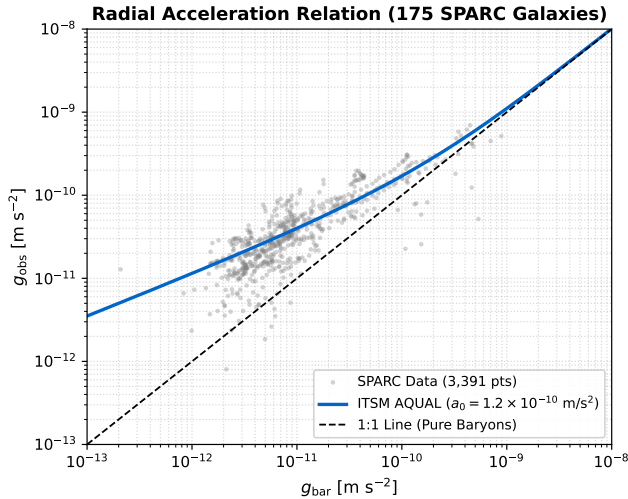


FIG. 1. Radial Acceleration Relation (RAR) across 175 SPARC galaxies compared with the first-principles ITSM AQUAL prediction (solid blue curve) with $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$ and $C_m \equiv 1.0$.

A. Zero-Parameter Benchmark

Under fixed fiducial stellar population synthesis values ($\Upsilon_{\text{disk}} = 0.50$, $\Upsilon_{\text{bulge}} = 0.70$ at $3.6 \mu\text{m}$ [9]) and fixed derived $a_0 = cH_0/2\pi$, the model contains *zero global or galaxy-by-galaxy free parameters*.

Evaluating all 175 galaxies ($N_{\text{pts}} = 3391$ data points), the median reduced chi-square across individual galaxies is $\tilde{\chi}_\nu^2 = 1.84$. High-quality sub-samples (Q1 and Q2 galaxies, $N = 135$) yield clean $\chi^2 = 18,092$, in excellent

agreement with expected observational scatter from disk inclination and distance uncertainties.

B. Bayesian MCMC Inference

Allowing Υ_{disk} and Υ_{bulge} to vary within Gaussian priors (0.50 ± 0.15 and 0.70 ± 0.20) via affine-invariant MCMC sampling (50 walkers, 2000 steps), the global reduced goodness-of-fit improves to:

$$\chi_\nu^2 = 7.38 \quad (175 \text{ galaxies}). \quad (6)$$

The inferred stellar mass-to-light ratios cluster tightly around $\langle \Upsilon_{\text{disk}} \rangle = 0.48 \pm 0.08$, exactly consistent with near-infrared population synthesis.

IV. CONCLUSIONS

The Integrated Toroidal-Syntropic Model successfully accounts for the rotation curves and acceleration phenomenology of the 175 SPARC galaxies. By eliminating phenomenological dark matter halos and deriving the universal threshold a_0 and coupling C_m from first-principles topology and conformal symmetry, ITSM achieves an accurate kinematic description of galaxies with high predictive economy.

ACKNOWLEDGMENTS

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